Experimental Studies and Constitutive Modeling of Static Liquefaction Instability in Sand–Clay Mixtures

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Experimental Studies and Constitutive Modeling of Static Liquefaction Instability in Sand–Clay Mixtures

Xilin Lü, A.M.ASCE[1]; Dawei Xue[2]; Bin Zhang[3]; and Qifeng Zhong[4]

Abstract: This paper examines static liquefaction phenomena in sand–clay mixtures from both experimental and theoretical perspectives. Consolidated undrained triaxial tests (CU tests) are implemented on a mixture of sand and clay to study the effects of initial relative density, confining pressure, and clay content on static liquefaction responses. By using the equivalent granular state parameter, a state-dependent hardening plasticity model is proposed to reproduce mechanical responses, as well as to replicate the observed static liquefaction phenomenon. In accordance with the second-order work theory, the criteria for predicting the potential instability and static liquefaction of sand–clay mixtures are obtained. The analyses show that (1) static liquefaction is prone to be promoted by low initial relative density and confining pressure; (2) even a small amount of clay particles can significantly increase the liquefaction susceptibility in loose sand–clay mixtures; (3) the proposed constitutive model can provide a satisfactory match between the test data and the simulations; and (4) even when the state of the soil is potentially unstable, soils can still remain stable unless second-order work vanishes, which indicates the inception of static liquefaction. **DOI: 10.1061/(ASCE)GM.1943-5622.0002472**.

Author keywords: Sand–clay mixtures; Consolidated undrained triaxial tests; Static liquefaction; State-dependent plasticity; Second-order work.

Introduction

Static liquefaction is a phenomenon related to a significant reduction in effective stress; it is accompanied by excessive strain and a buildup of pore-water pressure when saturated granular soils are subjected to undrained static loading (Yamamuro and Lade 1997; Vaid and Sivathayalan 2000; Mroz et al. 2003; Sabbar et al. 2017). This solid–fluid interaction–induced instability is commonly considered as a trigger for the failure of loose granular assemblies, e.g., the liquefaction-related failure of tailings dams (Fourie et al. 2001), slopes (Beddoe and Take 2015), and underground excavation faces (Zeng et al. 2019).

Numerous laboratory studies have been implemented to study the characteristics of static liquefaction in sands subjected to undrained loading (Castro 1969; Leong et al. 2000; Been and Jefferies 1985, 2004; Lashkari et al. 2017). These studies have contributed fundamental understandings of static liquefaction behavior. For example, when static liquefaction is activated, the stress–strain curve will peak at a small strain and then drop dramatically in the post-yielding stage until an ultimate state is reached at large strains. Such an ultimate state is referred to as the steady state, which is consistent with the critical state, and it is unique for the given sands (Poulos 1981; Chu 1995). However, natural sands are often not pure and contain some amount of fine particles of clay or silt (Xiao et al. 2019a, b; Ma et al. 2021). From the observations made from fields and labs, the existence of fine grains, even with a proportion of a value less than 10%, can play a nonnegligible role in the ultimate states and the static liquefaction behavior of sands (Yamamuro and Lade 1998; Rahman et al. 2014a; Dai et al. 2015, 2019; Xiao et al. 2017b; Krim et al. 2019). To reveal the effects provided by fine grains, many laboratory tests have been implemented. Amini and Qi (2000) have found that the antiliquefaction ability of sand–fines mixtures monotonically increases with the fines content, while an opposite conclusion has been made by Singh (1996) and Zlatovic and Ishihara (1997). Later studies (Ling 1992; Bouferra and Shahrour 2004; Choo and Burns 2015) have provided a possible explanation for these conflicting results. These studies have found that there exists a critical value for clay content (or fine grain content) that can reverse the relationship between the liquefaction resistance and the clay content. When the clay content is less than the critical value, fine particles can roll into the sand void upon shearing. Such processes can promote specimen shrinkage and an increase in pore fluid pressure, causing liquefaction instability. On the contrary, for sands with clay content larger than the critical value, small fine particles can fill up the void space between large sand particles, which makes it difficult for specimens to shrink. In this scenario, sand–clay mixtures become hard, making it difficult to liquefy them (Terzaghi and Peck 1996; Yamamuro and Lade 1997). Benahmed et al. (2015) have conducted a series of triaxial tests under undrained conditions and revealed that the critical fine content for the soils being used is about 20%. As noted by Zuo and Baudet (2015), the critical fine content varies from 20% to 50%, depending on the specific measuring methods used (Dash et al. 2010). The laboratory work done by Kim et al. (2018) shows that the shear strength analysis of direct shear tests might be used to estimate critical clay content for sand–clay mixtures. Huang and Zhao (2018) regarded the existence of small enough fines as improving liquefaction resistance due to

[1]Professor, Key Laboratory of Geotechnical and Underground Engineering of Ministry of Education and Dept. of Geotechnical Engineering, Tongji Univ., Shanghai 200092, China (corresponding author). Email: xilinlu@tongji.edu.cn

[2]Ph.D. Candidate, Dept. of Geotechnical Engineering, Tongji Univ., Shanghai 200092, China. Email: 1278641324@qq.com

[3]Graduate Student, Dept. of Geotechnical Engineering, Tongji Univ., Shanghai 200092, China. Email: 1069424403@qq.com

[4]Graduate Student, Dept. of Geotechnical Engineering, Tongji Univ., Shanghai 200092, China. ORCID: https://orcid.org/0000-0002-7850-8125. Email: 1932389@tongji.edu.cn

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the strong bond between particles and hydration adsorption of nano-suspension.

The theoretical prediction of static liquefaction is mainly based on the second-order work criterion both for the macroscopic scale (Lade et al. 1988; Buscarnera and de Prisco 2012; Buscarnera and Whittle 2013; Lü and Huang 2015) and for the microscopic scale (Zeng et al. 2019; Wu et al. 2020). The second-order work criterion for the macroscopic scale typically needs to be combined with an analytical constitutive model from critical state theory. Critical state theory for soils has been widely studied for both clays (Roscoe et al. 1958; Hamidi et al. 2015) and sands (Li and Wang 1998; Li and Dafalias 2000; Xiao et al. 2017a); the latter usually requires a state parameter to capture their mechanical behavior. Based on the use of a state parameter, Lü et al. (2017) have proposed a strain-hardening elastoplastic model for capturing the initiation of static liquefaction in cross-anisotropic sands under the multiaxial stress conditions. This constitutive framework has been extended by Lü et al. (2018) to simulate mechanical response and liquefaction instability in unsaturated sand with bubbles. Recent studies (Rahman et al. 2011, 2014b; Goudarzy et al. 2016) show that the use of the equivalent granular void ratio, in which the concept of threshold fine particle content is incorporated, can provide a possible way to formulate an equivalent state parameter for sand–fines mixtures, and, thereby, strengthen the predictions of mechanical behavior and liquefaction analyses.

This paper studies static liquefaction phenomena in sand–clay mixtures by conducting consolidated undrained triaxial tests and constitutive model–based theoretical analyses. The specimen used in the test is a mixture of sand and clay. The concept of the equivalent granular void ratio is introduced, and a state-dependent Mohr–Coulomb hardening plasticity model is proposed. Based on the second-order work theory, the criteria for predicting the potential instability and static liquefaction are derived. The model performance is assessed using the reported test data.

Experimental Studies

Materials and Experimental Setup

The specimens were prepared by mixing sand and clay. The main composition of the tested sand is silica (98.23%) and aluminum oxide (1.21%). Fig. 1 shows the grain size distributions of both materials. The physical properties of sand, clay, and their mixtures are presented in Tables 1–3, respectively. The specimens were shaped in the form of a cylinder with a diameter of 38 mm and a height of 76 mm. Then, consolidated undrained triaxial shear tests (e.g., CU tests) were conducted. The strain-controlled loading rate was 0.05%/min. The detailed experimental procedure is presented as follows:

1. Drying: The materials were dried in a drying oven at a temperature of 105°C for more than 24 h.
2. Installation: The specimens were prepared using the moist-tamping method. The dried sands and clay were mixed using different mass ratios and were then stirred together with water until no obvious clay agglomerates could be found. Then, the mixtures were placed in a sealed bag for more than 24 h. Before the sand–clay mixtures were taken out from the sealed bag, the water content of the samples located at different positions was measured to ensure that the spatial difference does not exceed 1%. The specimens were prepared in five layers (the mass of each layer is equal). The weight of each layer of soil sample was determined according to the desired void ratio (due to its relationship with initial density) and the water content. The surface of each layer was smoothed to avoid the formation of weak contact surfaces. To create a loose specimen, each layer of specimen was stirred to be loose enough.
3. Saturation: Carbon dioxide was imported from the specimen bottom to replace the air inside. This process lasted about two hours. Then, water (5 times the volume of the specimen) was imported from the bottom and drained out from the top boundary. A back-pressure saturation method was utilized to ensure a large enough B value ($\geq$0.98); the back pressures for all samples were 200 kPa.
4. Isotropic consolidation and undrained shear: The confining pressures of 30,100, and 200 kPa were applied. Consolidated undrained shear loading was then implemented. The test ended when the axial strain reached 20%.

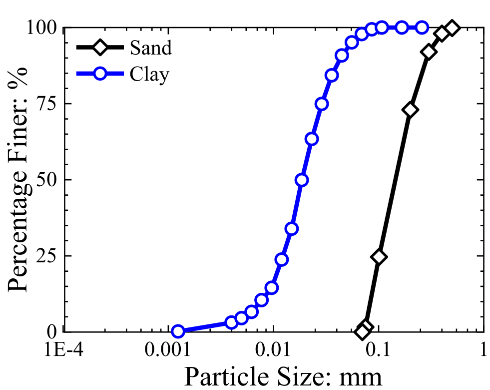

Fig. 1. Grain size distribution of sand and clay.

Test Results of Pure Sands

Before we investigate responses of sand–clay mixtures, we show the test results for pure sands, i.e., the stress paths, stress–strain curves, and pore pressure–strain curves, in Fig. 2. Four initial relative densities ($D_r = 1\%$, 10%, 50%, and 70%) and three confining pressures ($\sigma_c = 30$,100, and 200 kPa) were chosen for the tests. Fig. 2(a–c) shows the results obtained from very loose sands

Table 1. Physical properties of sand

Material	Specific gravity, G_s	Maximum void ratio, $e_{\max}$	Minimum void ratio, $e_{\min}$	Uniformity coefficient, C_u	Mean grain size, d_{50}
Sand	2.656	0.973	0.634	1.57	0.17

Table 2. Physical properties of clay

Material	Specific gravity, G_s	Liquid limit, W_L	Plastic limit, W_P	Plasticity index, I_P	Mean grain size, d_{50}
Clay	2.751	33.6%	15.8%	17.8%	0.019

Table 3. Physical properties of sand–clay mixtures

Sand–clay mixtures (%)	Specific gravity, G_s	Maximum void ratio, $e_{\max}$	Minimum void ratio, $e_{\min}$	Maximum dry density, $\rho_{\max}$	Minimum dry density, $\rho_{\min}$
$f_c = 5$	2.661	0.999	0.644	1.619	1.331
$f_c = 10$	2.667	1.023	0.573	1.696	1.318
$f_c = 15$	2.671	1.071	0.516	1.762	1.290

Note: f_c = clay content.

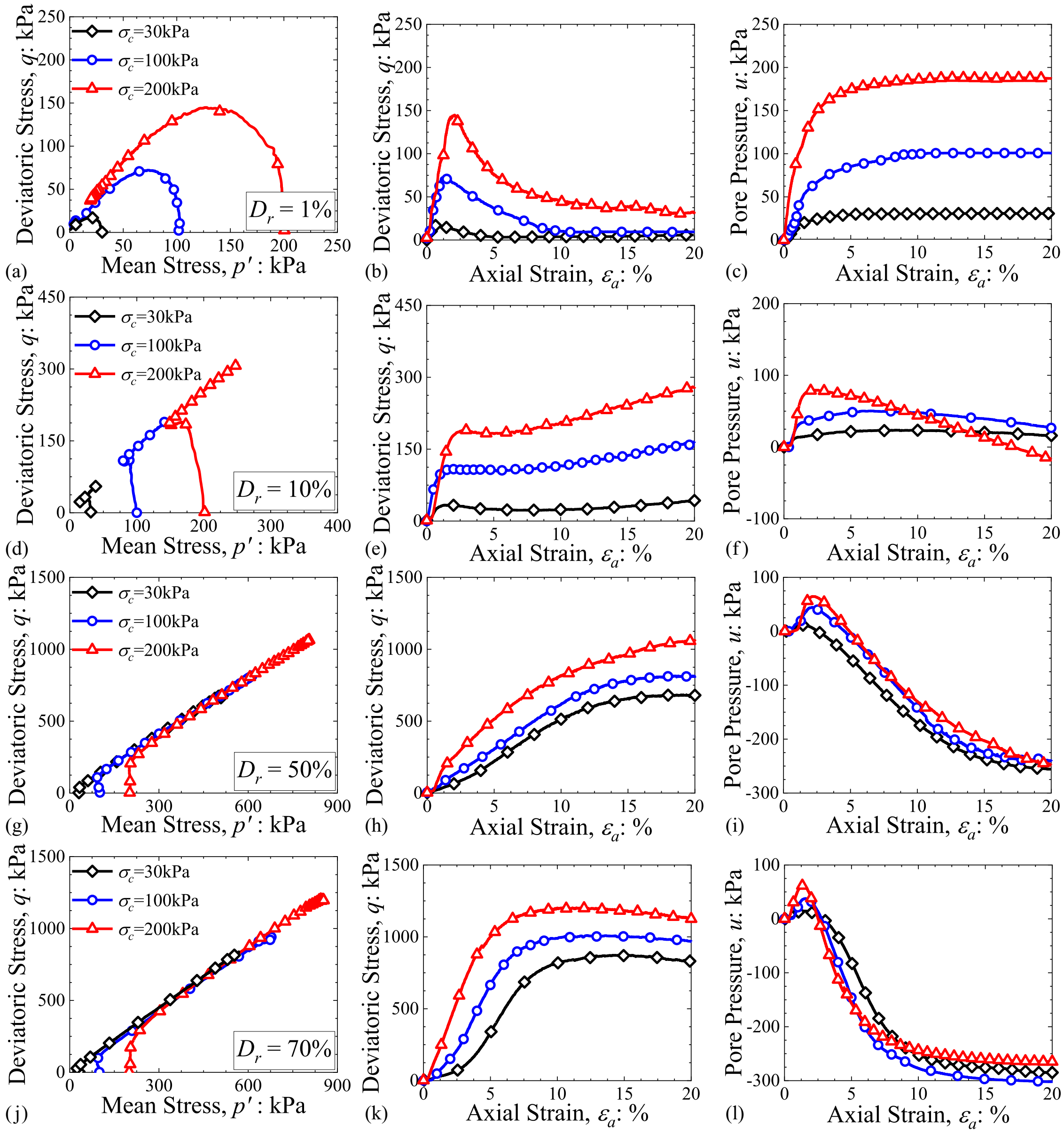

Fig. 2. Stress paths, stress–strain relations, and pore pressure–strain curves obtained from CU tests for pure sands (i.e., clay content $fc = 0\%$): (a–c) results of very loose sand, i.e., $Dr = 1\%$; (d–f) results of loose sand, i.e., $Dr = 10\%$; (g–i) results of medium dense sand, i.e., $Dr = 50\%$; and (j–l) results of dense sand, i.e., $Dr = 70\%$.

(i.e., $D_r = 1\%$). As the confining pressure increases from 30 kPa to 200 kPa, the stress paths, deviatoric stress curves, and pore-water pressure curves move upward. For $\sigma_c = 30$ and 100 kPa, the deviatoric stress reaches a peak and then gradually decreases to the same residual value (close to 0 kPa). Correspondingly, the pore-water pressure increases continuously upon loading, and finally approximately reaches the value of the applied confining pressures. This readily apparent phenomenon indicates an occurrence of static liquefaction. The confining pressure of 200 kPa, however, results in a different outcome. Specifically, residual deviatoric stress is nonzero. In addition, the pore-water pressure did not reach 200 kPa even at the end of loading, which illustrates the inhibition of high confining pressure on static liquefaction. When $D_r = 10\%$ is used [Figs. 2(d–f)], for all confining pressures, a slight strain-softening phase followed by a strain-hardening stage can be observed in the stress paths and stress–strain curves. This temporary strain-softening behavior indicates an ephemeral liquefaction phenomenon. The pore-water pressure for $\sigma_c = 30$ kPa increases monotonically, while $\sigma_c =$ 100 and 200 kPa result in a decreasing trend of pore-water pressure after the monotonically increased phase. When larger initial relative densities ($D_r = 50\%$ and 70%) are selected for CU tests, the stress paths and stress–strain curves for all cases always present a strain-hardening behavior, and liquefaction cannot occur [Figs. 2(g–l)]. The pore-water pressures for all cases increase at the beginning and then drop dramatically to negative values as loading proceeds. It is readily apparent that dense sands are less prone to static liquefaction.

Test Results of Sand–Clay Mixtures

In this section, the CU test results of sand–clay mixtures are presented. In all tests, a confining pressure of 30 kPa is used to better induce static liquefaction. Figs. 3(a–c) show the stress paths, stress–strain curves, and pore pressure–strain curves for different clay contents, f_c. The initial density D_r is set to 10%. When the

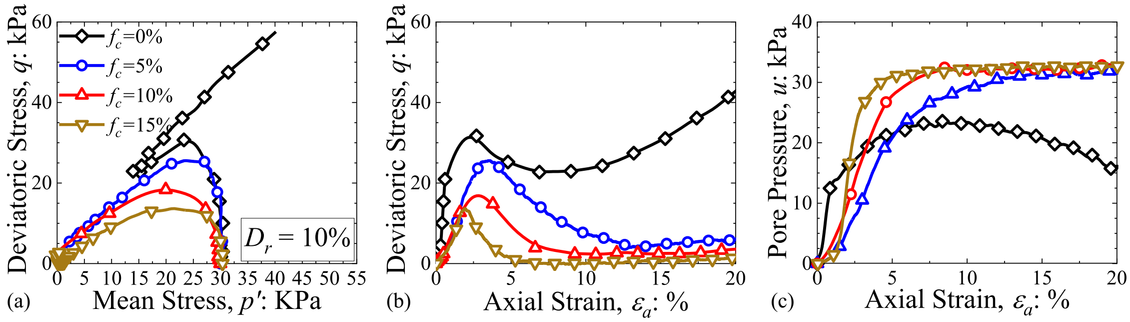

Fig. 3. CU test results obtained from a loose specimen ($Dr = 10\%$) with different clay contents fc (initial confining pressure $\sigma_c = 30$ kPa): (a) stress paths; (b) stress–strain relations; and (c) pore pressure–strain curves.

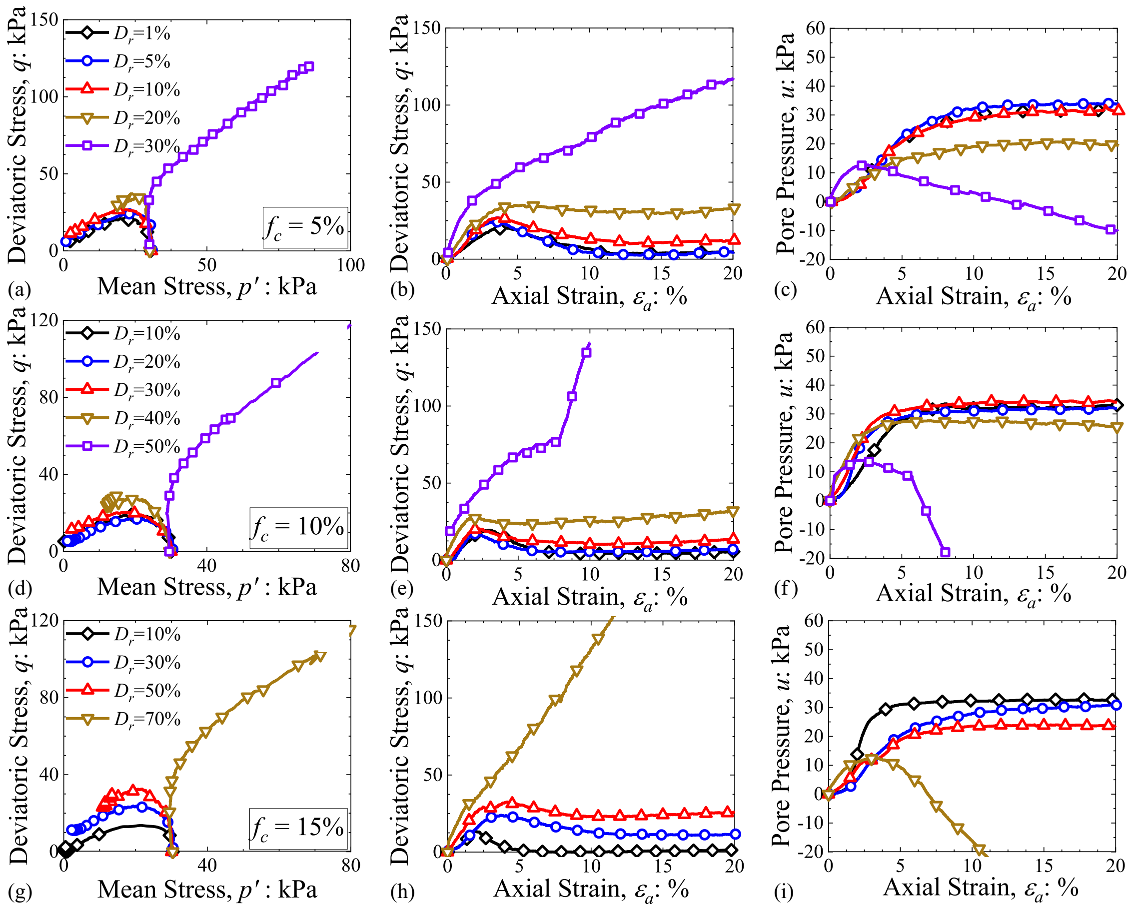

Fig. 4. Stress paths, stress–strain relations, and pore pressure–strain curves obtained from different initial relative densities Dr based on constant clay content fc (initial confining pressure $\sigma_c = 30$ kPa): (a–c) results of $fc = 5\%$; (d–f) results of $fc = 10\%$; and (g–i) results of $fc = 15\%$.

clay content reaches a value greater than 5%, a complete static liquefaction phenomenon can be observed, while pure sands show temporary liquefaction (i.e., a strain-softening phase followed by a strain-hardening stage). The increase in clay content makes the specimen more conducive to static liquefaction.

To better show the promotion effects of clay particles to static liquefaction in sand–clay mixtures, Fig. 4 shows the CU test results obtained from different initial relative densities based on constant clay contents. When $f_c = 5\%$, D_r should be more than 30% to inhibit static liquefaction. By contrast, as f_c increases up to 15%, the critical D_r needed for preventing static liquefaction can be as large as 70%. This observation can provide a guideline to geotechnical engineering that a small clay content can dramatically reduce the resistance of a sand–clay mixture to static liquefaction instability, which challenges the safety of engineering processes.

A possible explanation of how fine content impacts the mechanical responses and static liquefaction can be as follows: when the clay content is less than the critical value, upon loading, clay particles can slide into the void between sand particles, which means the specimens are prone to shrinkage. As a result, pore fluid pressure increases, and effective stress decreases, which promotes the inception of static liquefaction and strain-softening behavior. However, with an increase in clay content, clay particles can fill up the void space between solids and form a denser state. Therefore, specimens are not prone to shrinkage, and, thus, static liquefaction might be suppressed.

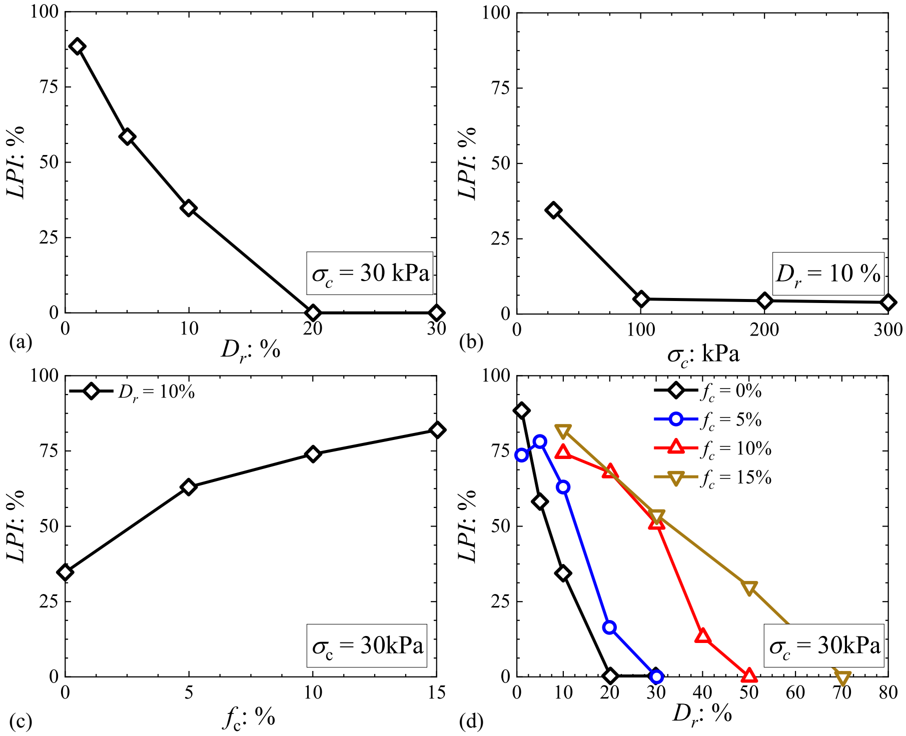

Fig. 5. LPI analysis: (a) LPI versus *Dr* for pure sands; (b) LPI versus *σc* for pure sands; (c) LPI versus *fc* for sand–clay mixtures; and (d) LPI versus *Dr* for sand–clay mixtures.

Analysis of Static Liquefaction Potential

Static liquefaction potential analysis was enclosed through computations of the liquefaction potential index (LPI) (Ng et al. 2004), given by

$$LPI = \frac{q_{\text{peak}} - q_{qss}}{q_{\text{peak}}} \tag{1}$$

where q_{peak} and q_{qss} are the deviatoric stress values at peak and quasi-steady state, respectively. Eq. (1) can be used to quantitatively assess static liquefaction potential in experimental results. $\text{LPI} = 0$ indicates that no strain-softening behavior, as well as liquefaction phenomenon, is activated, while $\text{LPI} = 1$ is recognized as a symbol of complete liquefaction instability. In contrast, $0 < \text{LPI} < 1$ usually represents a medium state between $\text{LPI} = 0$ and $\text{LPI} = 1$.

Fig. 5 shows the LPI analysis for pure sands and sand–clay mixtures. It can be observed from Figs. 5(a and b) that LPI decreases as the initial relative density D_r and confining pressure σ_c increases, indicating a suppressed trend of static liquefaction. The effects of clay content f_c on liquefaction potential are shown in Fig. 5(c). Under a loose state, the existence of clay particles minifies the liquefaction resistance of specimens. As clay content f_c increases, PLI increases monotonically. Fig. 5(d) shows the relationships between PLI and initial relative density D_r under different clay contents f_c. When f_c increases from 0% to 15%, the critical D_r required for obtaining a complete liquefaction resistance significantly increases from 20% to 70%. These obtained results are consistent with those observed in previous sections.

Constitutive Model

To reproduce the mechanical responses, as well as to theoretically predict the static liquefaction instability in sand–clay mixtures, a state-dependent constitutive model is proposed as a basis in this section. The equivalent granular state parameter is utilized to consider the effects of clay particles on the mechanical behavior of sand–clay mixtures. The criteria obtained from second-order work theory for detecting potential instability and static liquefaction instability are discussed.

Equivalent Granular State Parameter

To formulate a state-dependent plasticity model for sands, a state parameter ψ defined by Been and Jefferies (1985) should be used to determine the state of soils ($\psi < 0$ represents a dilative state, while $\psi > 0$ indicates a contractive state). This parameter is formulated using the difference between the current void ratio and the void ratio at the critical state under the same mean pressure state, given by

$$\psi = e - e_c \tag{2}$$

where e_c = void ratio at the critical state line, expressed as follows:

$$e_c = e_{c0} - \lambda_c \left(\frac{p'}{p_a}\right)^{\xi} \tag{3}$$

where $p_a = 101.3$ kPa is the atmospheric pressure; $p' = \sigma_{ii}/3 - u$ = effective mean pressure; u = pore pressure; and e_{c0}, λ_c, and ξ = material parameters used to determine the critical state line in the (e, p') plane.

The use of ψ leads to a good match between numerical simulations and experimental results for pure sands. For sands with fine particles (e.g., sand–clay mixtures), however, the simulation and experimental results did not match well (Lashkari 2016; Yin et al. 2016). The critical state line for base granular material can be altered by fine particles (Tong et al. 2022). The effects on mechanical behavior provided by fines can be augmented when particle breakage emerges (Tong et al. 2018). Fine particle migration is also a nonnegligible microscale process, in that it can affect the macroscale mechanical behavior (Zhang et al. 2021). The equivalent granular state parameter ψ^* based on the equivalent granular void ratio e^* (Rahman et al. 2011, 2014a; b) is an alternative for better capturing mechanical responses in sand–clay mixtures. This equivalent granular void ratio e^* is expressed as follows:

$$e^* = \frac{e + (1+b)f_c}{e - (1-b)f_c} \tag{4}$$

where b = fraction of fine particles that are active in the force structure, and is between 0 and 1, and can be expressed as follows:

$$b = \left[1 - \exp\left(-\mu \frac{(f_c/f_{\mathrm{thre}})^{n_b}}{k}\right)\right]\left(r\frac{f_c}{f_{\mathrm{thre}}}\right)^r \tag{5}$$

where $k = 1 - r^{0.25}$, $r = 1/\chi$, $\chi = D_{10}/d_{50}$, and μ = a fitting parameter which is chosen as 0.30, f_{thre} is referred to as a threshold clay content. Specifically, there exists a critical clay content which leads to the following phenomenon: as f_c increases, the liquefaction possibility of sand–clay mixtures increases when $f_c < f_{\mathrm{thre}}$ but decreases when $f_c \geq f_{\mathrm{thre}}$. It should be noted here that the use of Eq. (4) requires $f_c < f_{\mathrm{thre}}$, which means that this paper will not discuss simulations for the CU tests with $f_c \geq f_{\mathrm{thre}}$. Based on experimental data analyses, Rahman et al. (2008) have proposed a computation method for f_{thre}, given by

$$f_{\mathrm{thre}} = 0.4\left(\frac{1}{1+e^{\alpha - \beta\chi}} + \frac{1}{\chi}\right) \tag{6}$$

where α and β = fitting parameters. Here, the values of α and β are 0.5 and 0.13, the value of f_{thre} is 29.6%.

Using the aforementioned, the equivalent granular state parameter ψ^* can be defined as follows:

$$\psi^* = e^* - e_c^* \tag{7}$$

where e_c^* = equivalent void ratio at the equivalent granular critical state line, expressed as follows:

$$e_c^* = e_{c0}^* - \lambda_c^*\left(\frac{p'}{p_a}\right)^{\xi^*} \tag{8}$$

where e_{c0}^*, λ_c^*, and ξ^* = material parameters for equivalent granular critical state line in the (e^*, p') plane.

Figs. 6(a and b) show the critical state lines and equivalent critical state line for different clay contents f_c, respectively. The parameters are given in Tables 4 and 5. It can be observed that the critical state line moves downward as f_c increases. In the meantime, the inclination of the critical state line also becomes larger. By contrast, sand–clay mixtures with different f_c always show the same equivalent granular critical state line. Recent experiments done by Goudarzy et al. (2021) indicate that the fine type can perturb the alteration of the critical state line slope under different fine contents. Their analysis suggested that the physics behind such phenomenon may lie in different bonding structures between the sand particles caused by fines, which means that the use of different fine types can lead to differences (i.e., critical state line slope variations) in the test results in this work to that reported in Goudarzy et al. (2021). This aspect is still not fully understood in the research community and remains an interesting research line. However, regardless of fine type, the critical state line generally moves downward with increasing clay content (Goudarzy et al. 2021), which is consistent with that observed in this work.

It is worth mentioning that recent studies (Yang et al. 2015) have examined the rationale of the concepts of the equivalent granular void ratio and the parameter b, and the limitations. They pointed out that the rationale behind the equivalent granular void ratio, particularly the physical meaning of the parameter b involved in the definition, remains open for discussion. For the expressions here used in Eqs. (5) and (6), the size disparity ratio is considered, and a satisfactory match between experimental data and simulations can be obtained, which is shown in the following sections. To better understand the physical rationale behind the equivalent granular void ratio and the parameter b, it is still a valid extension of this work in the future to further account for the rich effects provided by intergranular contacts.

State-Dependent Plasticity Model

The yield function is proposed on the basis of the widely used Mohr–Coulomb elastoplastic constitutive model,

Table 4. Material properties of the critical state line

Material/Parameters	$f_c = 0\%$	$f_c = 5\%$	$f_c = 10\%$	$f_c = 15\%$
e_{c0}	0.95	0.96	0.90	0.87
λ_c	0.039	0.079	0.112	0.151
ξ	0.7	0.7	0.7	0.7

Table 5. Material properties of the constitute model based on equivalent granular critical state line

Parameters	e_{c0}^*	λ_c^*	ξ^*	G_0	v	M_{cs}	n_p	n_d	d_0	A
Value	0.97	0.061	0.7	50	0.1	1.38	1	3	0.88	0.001

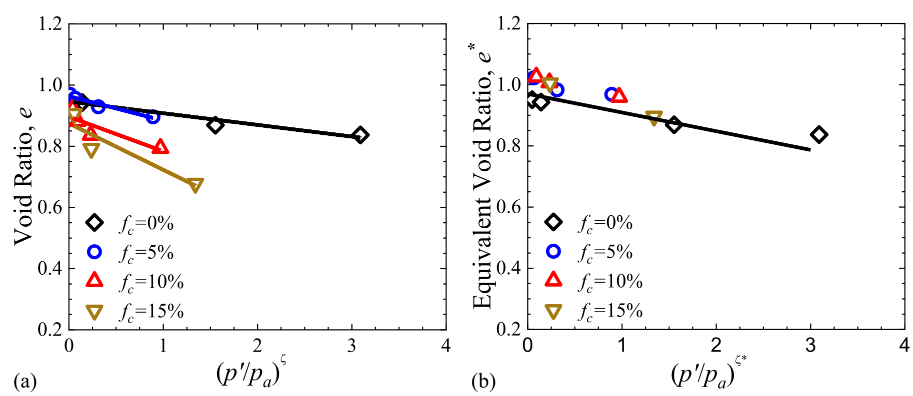

Fig. 6. Critical state analysis: (a) critical state lines obtained from different clay contents fc; and (b) equivalent critical state line for sand–clay mixtures.

expressed as follows:

$$F = q - Mp' = 0 \tag{9}$$

where $q = \sqrt{3s'_{ij}/s'_{ij}}$ = deviatoric stress, $s'_{ij} = \sigma'_{ij} - \delta_{ij}p'$; $\sigma'_{ij} = \sigma_{ij} - u$; δ_{ij} = Kronecker symbol; and M is used to describe the hardening behavior (Pietruszczak and Stolle 1987; Huang et al. 2010), assumed to follow the hyperbolic law:

$$M = M_p \frac{\varepsilon_s^p}{A + \varepsilon_s^p} \tag{10}$$

where $\varepsilon_s^p = \sqrt{2e_{ij}e_{ij}/3}$; $e_{ij} = \varepsilon_{ij} - \delta_{ij}\varepsilon_{kk}/3$; and A = fitting parameter.

M_p is assumed to be related to the equivalent soil state, given by

$$M_p = M_{cs} \exp(-n_p\psi^*) \tag{11}$$

where M_{cs} = stress ratio in the critical state line; and n_p = material parameter.

The dilatancy function can be expressed as follows:

$$D = \frac{d\varepsilon_v^p}{d\varepsilon_s^p} \tag{12}$$

where $d\varepsilon_v^p = d\varepsilon_{kk}^p$; $d\varepsilon_s^p = \sqrt{2de_{ij}^p de_{ij}^p/3}$; and $de_{ij}^p = \varepsilon_{ij}^p - \delta_{ij}d\varepsilon_{kk}^p/3$.

A framework is needed to define a unique relationship between the stress ratio and the dilatancy. To consider the effects of the material state, Manzari and Dafalias (1997) and Li and Dafalias (2000) have incorporated a state parameter into the dilatancy function. Here, the state parameter is replaced with the equivalent granular state parameter, expressed as follows:

$$D = A_d[M_d - M] = A_d[M_{cs}\exp(n_d\psi^*) - M] \tag{13}$$

where $A_d = d_0/M_{cs}$, d_0, and n_d are material parameters related to the peak stress ratio. For simplicity, $A_d = d_0/M_{cs}$ is assumed to be 1.0. As a result, Eq. (13) is the same as the dilatancy function used in the Cam-clay model. The plastic potential can be formulated as follows:

$$Q = q + M_d p' \ln\frac{p'}{p_a} = 0 \tag{14}$$

The model can be completed with a nonlinear elasticity, given by

$$\sigma_{ij} = D_{ijkl}^e \varepsilon_{kl}^e \tag{15}$$

$$D_{ijkl}^e = \left(\bar{K} - \frac{2}{3}\bar{G}\right)\delta_{ij}\delta_{kl} + \bar{G}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \tag{16}$$

where $\bar{K}$ and $\bar{G}$ = bulk modulus and shear modulus, respectively. These two parameters are related to the current equivalent void ratio, expressed as follows:

$$\begin{cases} \bar{K} = \dfrac{2(1+v)}{3(1-2v)}G \\ \bar{G} = G_0 p_a \dfrac{(2.97 - e^*)^2}{1 + e^*}\sqrt{\dfrac{p'}{p_a}} \end{cases} \tag{17}$$

where G_0 = regression constant of the elastic shear modulus; and v = Poisson's ratio. The rate form of the elastoplastic constitutive relationship can be formulated as follows:

$$\dot{\sigma}_{ij} = D_{ijkl}^{ep}\dot{\varepsilon}_{kl} \tag{18}$$

$$D_{ijkl}^{ep} = D_{ijkl}^e - D_{ijkl}^p = D_{ijkl}^e - \frac{D_{ijmn}^e \dfrac{\partial Q}{\partial \sigma_{mn}} \left(\dfrac{\partial F}{\partial \sigma_{pq}}\right)^T D_{pqkl}^e}{\left(\dfrac{\partial F}{\partial \sigma_{uv}}\right)^T D_{uvst}^e \dfrac{\partial Q}{\partial \sigma_{st}} + H_p} \tag{19}$$

where $H_p = -(\partial F/\partial M)(\partial M/\partial \varepsilon_s^p)$ is the hardening modulus.

To simulate the stress–strain responses of sand–clay mixtures under the triaxial condition conveniently, Eq. (19) can be rewritten in $p'-q$ form:

$$\begin{Bmatrix} \dot{p}' \\ \dot{q} \end{Bmatrix} = D_{pq}^{ep}\begin{Bmatrix} \dot{\varepsilon}_v \\ \dot{\varepsilon}_s \end{Bmatrix} \tag{20}$$

where D_{pq}^{ep} can be expressed as follows:

$$D_{pq}^{ep} = \begin{bmatrix} \bar{K} + \dfrac{\bar{K}^2 M(M_d - M)}{H_p + \bar{K}M(M - M_d) + 3\bar{G}} & \dfrac{-3\bar{K}\bar{G}(M_d - M)}{H_p + \bar{K}M(M - M_d) + 3\bar{G}} \\ \dfrac{3\bar{K}\bar{G}M}{H_p + \bar{K}M(M - M_d) + 3\bar{G}} & 3\bar{G} - \dfrac{9\bar{G}^2}{H_p + \bar{K}M(M - M_d) + 3\bar{G}} \end{bmatrix} \tag{21}$$

The equivalent granular state parameter can also be incorporated into generalized plasticity theories, as reported by Sun et al. (2021). In this work, we focused on the use of the traditional explicit yield function and plastic potential, in that traditional conceptions can be widely understood and accepted. Another novel work of constitutive modeling of sand–fines mixtures proposed by Wei and Yang (2019) did not directly use the concept of the equivalent granular state parameter. By contrast, they introduced the state parameter dependency of the elastic shear modulus and a state-parameter-dependent hardening modulus, in which the effects of fine particles are considered according to experimental observations.

Criteria for Static Liquefaction Instability

According to the second-order work theory, material stability is maintained if the following conditions can be satisfied:

$$d^2w = d\boldsymbol{\sigma}':d\boldsymbol{\varepsilon} > 0, \forall\|d\boldsymbol{\varepsilon}\| \neq 0 \tag{22}$$

where d^2w = second-order work; $d\boldsymbol{\sigma}'$ = effective stress increment tensor; and $d\boldsymbol{\varepsilon}$ = strain increment tensor. If there is a stress state that violates Eq. (20) (i.e., $d^2w \leq 0$), the granular system of interest can be seen as potentially unstable. It should be noted that Eq. (22) is a sufficient condition for material stability, while the vanishing or negative values of second-order work (i.e., $d^2w \leq 0$) are a necessary but insufficient condition for liquefaction instability (Chu and Leong 2001). The answer to the question whether static liquefaction instability occurs also depends on the stress path. To determine the necessary and sufficient conditions for the occurrence of static liquefaction, Lade (1992) has proposed additional conditions, which can be expressed as follows:

$$\frac{\partial F}{\partial \sigma'_{ij}}\delta_{ij} < 0 \tag{23}$$

$$\frac{\partial Q}{\partial \sigma'_{ij}}\delta_{ij} > 0 \tag{24}$$

By observing Eqs. (9), (14), (23), and (24), it can be noticed that Eq. (23) can be fulfilled naturally, while Eq. (24) can be also fulfilled when $M_d - M > 0$. Therefore, the necessary and sufficient conditions for the static liquefaction instability in soils are (1) the volume of the soil reduces under undrained conditions; and (2) the second-order work becomes nonpositive (Lü and Huang 2015).

By looking into the condition under which the determinant of the symmetric part of the elastoplastic tensor vanishes or become negative, an potential instability index can be obtained. To this end, Eq. (18) is inserted into $d^2w = d\boldsymbol{\sigma}':d\boldsymbol{\varepsilon} \leq 0$. Then, we have

$$d\boldsymbol{\varepsilon}:\mathbf{D}^{ep}:d\boldsymbol{\varepsilon} \leq 0, \ \forall\|d\boldsymbol{\varepsilon}\| \neq 0 \tag{25}$$

Eq. (25) indicates the vanishing determinant value of the symmetric part of $\mathbf{D}^{ep}$, i.e., $\det(\mathbf{D}^{ep}_{sys}) \leq 0$. Considering a triaxial condition, the criteria for potential instability can be expressed as follows:

$$\det[(\mathbf{D}^{ep}_{pq})_{sys}] \leq 0 \tag{26}$$

Eq. (26) is a necessary but insufficient condition of Eq. (25). When Eq. (26) is fulfilled, the granular system enters a potentially unstable region. But whether the system becomes unstable also depends on specific stress paths. The soil assembly can stay stable, even if Eq. (26) is satisfied.

In the following section, the constitutive model and the criteria are used to reproduce mechanical responses, as well as to detect the stability of the specimens tested in the previous sections.

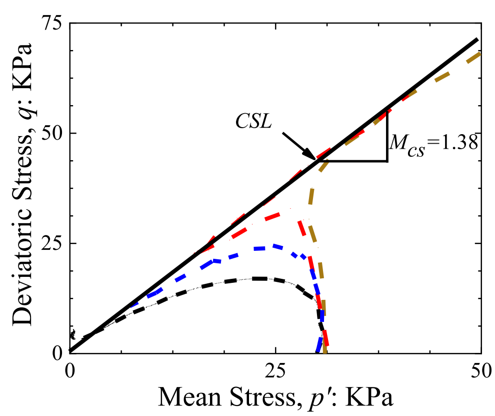

Fig. 7. Critical state line of pure sands in the $p'-q$ plane.

Model Prediction

Model Calibration

There are 10 parameters, namely, f_c, b, f_{thre}, G_0, M_{cs}, v, n_p, n_d, d_0, and A, that need to be determined for simulations. The physical relevance of parameters f_c, f_{thre}, b, G_0, and v has been explained in previous sections. The clay content f_c can be obtained directly. The parameter b for different f_c is determined using Eq. (5). The threshold clay content f_{thre} can be computed according to Eq. (6). The parameter M_{cs} is the slope of the critical state line in the (p', q) plane (Fig. 7). G_0 and v are elastic parameters and can be calibrated by using Eq. (17). The parameters n_p, n_d, d_0, and A are fitting parameters and have no specific physical meanings. They cannot be determined directly from existing test results and will be calibrated by fitting against experimental data. These calibrated parameters are given in Table 5. The comparisons between test data and simulations are shown in the following subsections.

Numerical Simulations and Liquefaction Analyses

Simulations of Pure Sands

Simulations of pure sands are shown in Figs. 8–11. Fig. 8 shows the simulations of CU tests for pure sands with different confining pressures. The sand state is very loose ($D_r = 1\%$). The simulated mechanical responses can match well with the experimental data. Static liquefaction is predicted using the second-order work, d^2w, the vanish of which is marked by black forks. Besides, $\det[(\mathbf{D}^{ep}_{pq})_{sys}] \leq 0$ is used to predict the onset of potential instability, and it is marked by green forks. Fig. 9 shows the tracking of state variables. It can be seen from Fig. 9(a) that when $\det[(\mathbf{D}^{ep}_{pq})_{sys}]$ becomes negative, the stress states for all confining pressures enter a potentially

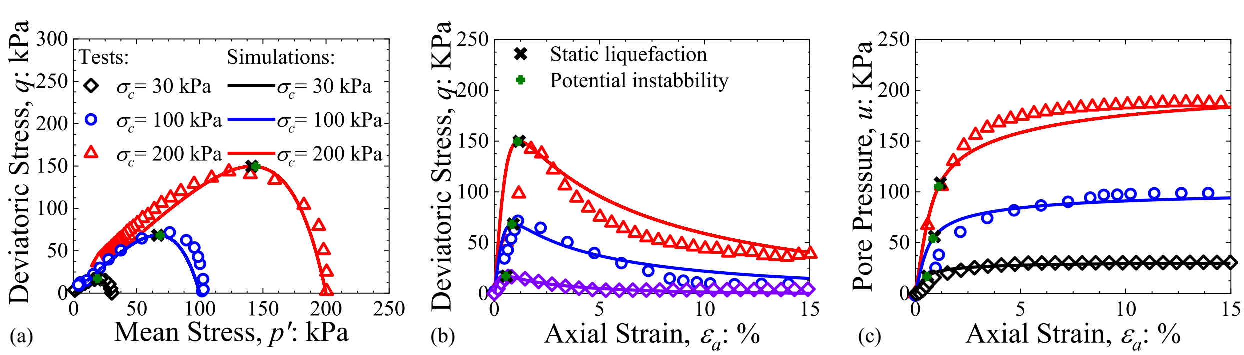

Fig. 8. Simulations of CU tests obtained from different initial confining pressures σc (initial relative density Dr = 1%; clay content fc = 0%): (a) stress paths; (b) stress–strain relations; and (c) pore pressure–strain curves.

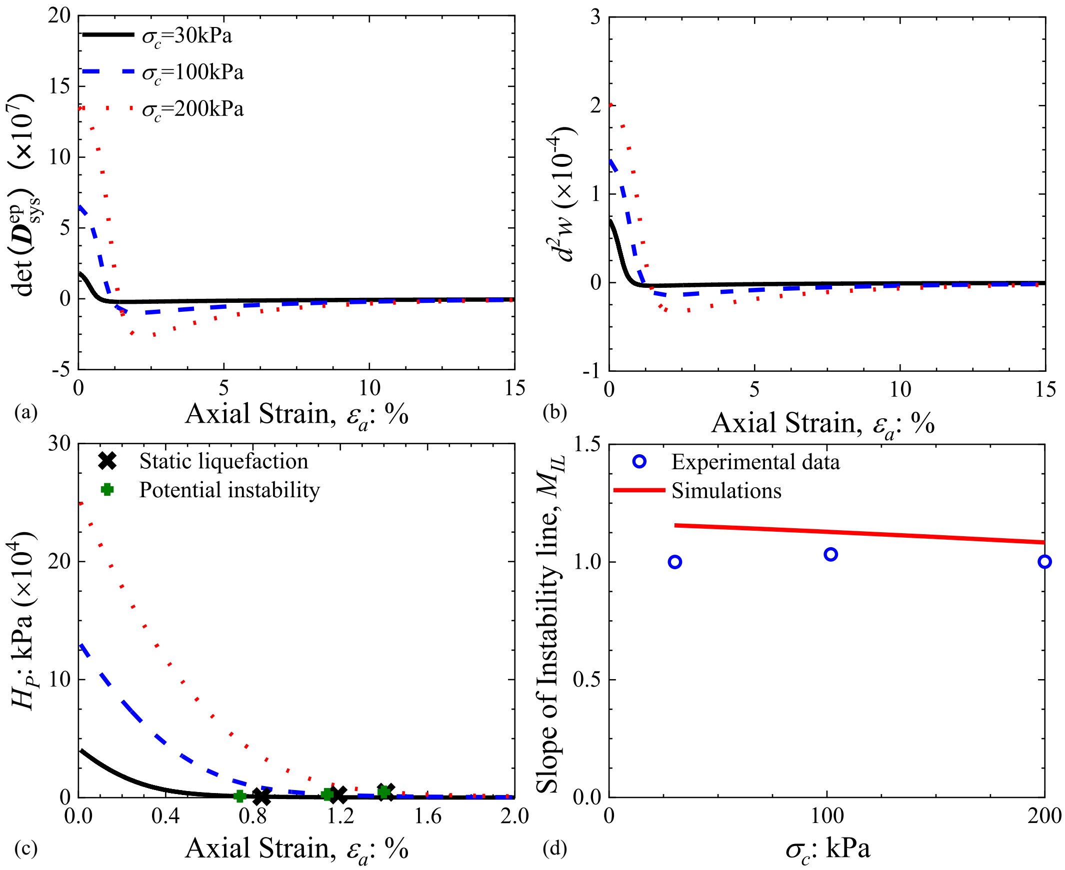

Fig. 9. Evolution of state variables in numerical simulations (initial relative density $Dr = 1\%$; clay content $fc = 0\%$): (a) determinant of the symmetric part of the elastoplastic tensor; (b) second-order work; (c) hardening modulus; and (d) instability line slope.

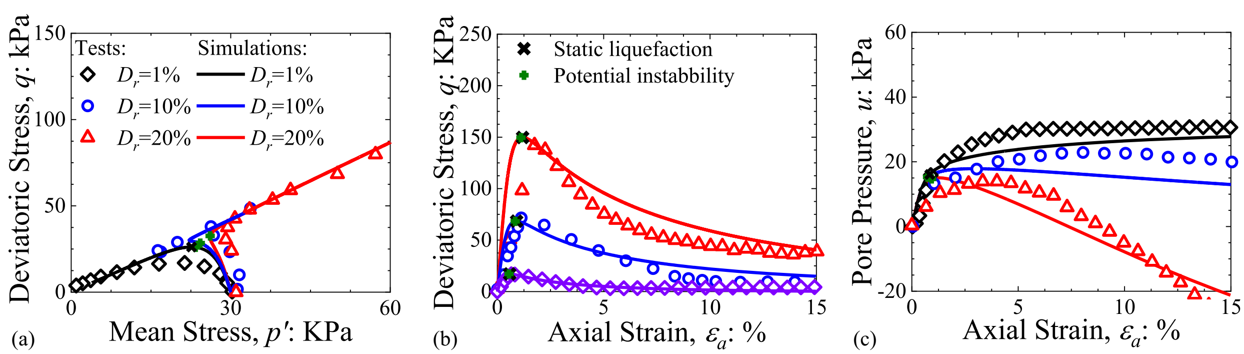

Fig. 10. Simulations of CU tests obtained from different initial relative densities Dr (initial confining pressure $\sigma c = 30$ kPa; clay content $fc = 0\%$): (a) stress paths; (b) stress–strain relations; and (c) pore pressure–strain curves.

unstable regime. Fig. 9(b) shows the evolution of second-order work. The simulations predict static liquefaction soon after the stress state entered the potentially unstable regime. The hardening modulus evolution curves reported in Fig. 9(c) show that the hardening modulus H_p is greater than 0 even when liquefaction occurs, indicating that the simulation is still in the hardening stage. Fig. 9(d) shows the evolution of the instability line slope. The simulations and test data match satisfactorily.

Fig. 10 shows the simulations of CU tests for pure sands with different initial densities. The confining pressure is chosen to be 30 kPa. With the increase in the relative density, the ability of sands to resist static liquefaction gradually increases. $D_r = 1\%$ results in complete static liquefaction, while $D_r = 20\%$ leads to no static liquefaction phenomenon. Fig. 11(a) shows the evolution of $\det[(\mathbf{D}^{ep}_{pq})_{sys}]$. All simulations predict $\det[(\mathbf{D}^{ep}_{pq})_{sys}] \leq 0$ at some points. Fig. 11(b) shows the second-order work evolution. It is observed that the simulation obtained from $D_r = 1\%$ predicts static liquefaction instability, i.e., $d^2w \leq 0$. For simulations with $D_r = 10\%$ and 20%, however, the second-order work is always positive and, therefore, no static liquefaction is predicted. Fig. 11(c) shows the hardening modulus evolution. The simulated curves are almost the same, indicating that the initial relative density has negligible effects on the hardening modulus H_p. The simulated

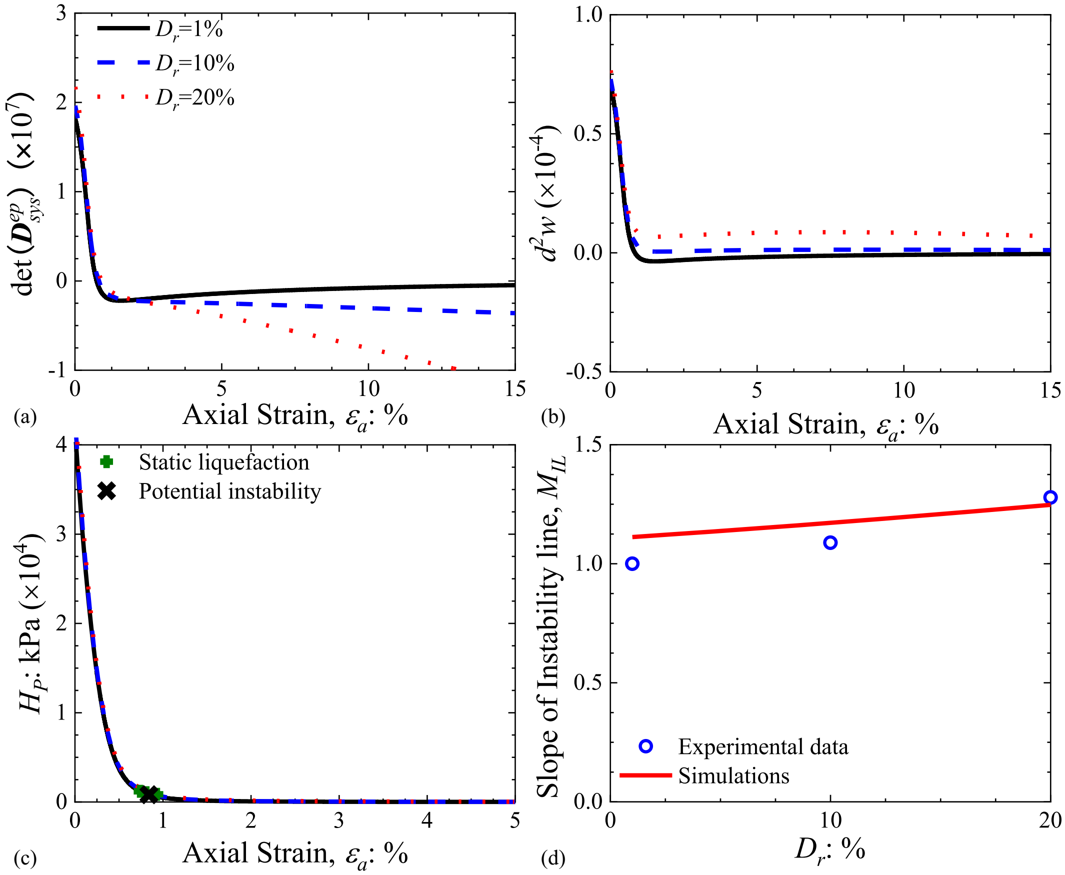

Fig. 11. Evolution of state variables in numerical simulations (CU tests: initial confining pressure $\sigma c = 30$ kPa; clay content $fc = 0\%$): (a) determinant of the symmetric part of the elastoplastic tensor; (b) second-order work; (c) hardening modulus; and (d) instability line slope.

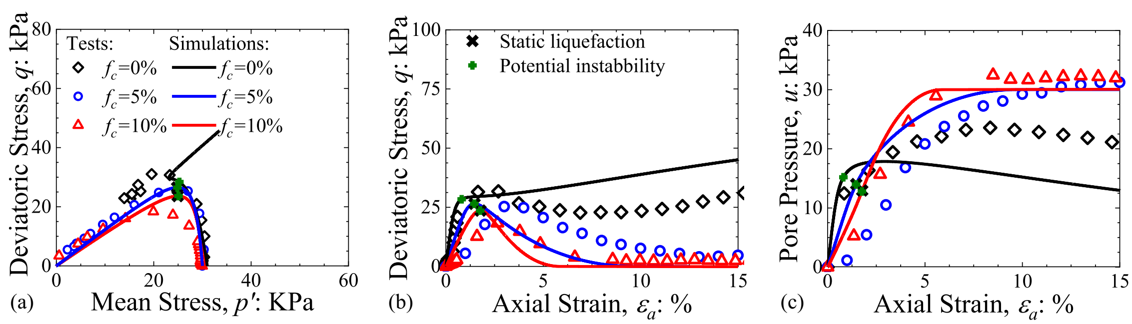

Fig. 12. Simulations of CU tests obtained from different clay contents fc (initial confining pressure $\sigma_c = 30$ kPa; initial relative density $Dr = 10\%$): (a) stress paths; (b) stress–strain relations; and (c) pore pressure–strain curves.

evolution of the instability line slope is shown in Fig. 11(d). The simulated result can match well with the experimental data.

~~Simulation~~ of Sand–Clay Mixtures

The CU tests of sand–clay mixtures with different clay contents are simulated and shown in Figs. 12 and 13. The initial confining pressure is chosen to be 30 kPa. Fig. 12 shows the mechanical responses obtained from tests and simulations. It is clearly shown that the simulations for $f_c = 0\%$ and 5% can well match the test data, while $f_c = 10\%$ leads to some acceptable mismatches. As f_c increases, the sand–clay mixtures underwent a transition from temporary liquefaction to complete static liquefaction. In the meantime, the residual deviatoric stress gradually decreases to 0. Fig. 13(a) shows the evolution of det$[(\mathbf{D}_{pq}^{ep})_{sys}]$. It can be observed that for all simulations, det$[(\mathbf{D}_{pq}^{ep})_{sys}]$ changes from positive to negative, indicating the possible occurrence of instability. The second-order work variations predict the occurrence of liquefaction for $f_c = 5\%$ and 10%, as shown in Fig. 13(b). The simulation obtained from $f_c = 0\%$, however, always shows a positive-valued second-order work evolution. The evolution of hardening modulus H_p is plotted in Fig. 13(c). As f_c increases, H_p moves downward. Fig. 13(d) shows the relationship between instability line slope and clay content f_c. When f_c increases, the slope of the instability line obtained from both tests and

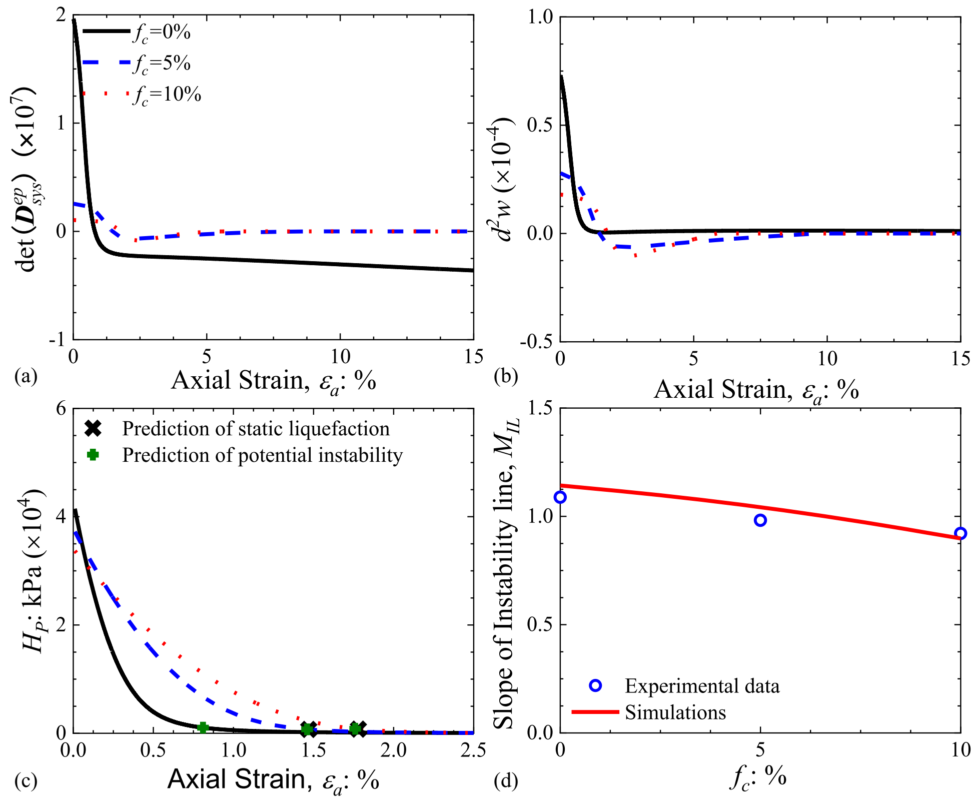

Fig. 13. Evolution of state variables in numerical simulations (CU tests: initial confining pressure $\sigma_c = 30$ kPa; initial relative density $Dr = 10\%$): (a) determinant of the symmetric part of the elastoplastic tensor; (b) second-order work; (c) hardening modulus; and (d) instability line slope.

simulations decreases slightly. It is readily apparent that the simulations and the experimental results match satisfactorily.

Conclusions

This paper presents a series of consolidated undrained triaxial tests (CU tests) on sand–clay mixtures. The effects of initial relative density, confining pressure, and clay content on static liquefaction behavior are analyzed. Based on the equivalent granular state parameter, a state-dependent hardening plasticity model is proposed to study the static liquefaction of sand–clay mixtures. The criteria for predicting the potentially unstable state and inception of static liquefaction are derived based on the second-order work theorem. The main conclusions of this paper can be given as follows:

a. The static liquefaction for pure sands is susceptible to the initial relative density and confining pressure. Liquefaction instability is inclined to occur once the initial relative density and confining pressure are relatively low.
b. For a loose sand–clay mixture, even a small amount of clay particles can raise the liquefaction susceptibility. As clay content increases, the minimum initial density required to inhibit static liquefaction increases significantly.
c. The use of the equivalent granular state parameter leads to a good match between the test data and the numerical simulations. The onset of static liquefaction predicted by second-order work also matches well with the test results.
d. By comparing the onset of static liquefaction predicted by second-order work and the potential instability predicted by the determinant of the symmetric part of the elastoplastic tensor, it can be found that when the material state is potentially unstable, soils can still remain stable unless second-order work shows a negative value.

Data Availability Statement

All data, models, and code that support the findings of this study are available from the corresponding author upon reasonable request.

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